

Skills Summary

Signs and Conjugates with Complex Numbers

Use this reference while completing your worksheet or when studying.

Skill 1 — Signs when you add, subtract, multiply

Two rules that fix most sign errors:

- A minus in front of a complex number hits *both* parts: $-(c - di) = -c + di$.
- $i^2 = -1$, so the last FOIL term always flips sign: $bi \cdot (-di) = -bd i^2 = +bd$.

Example: Multiply $(3 + 4i)(1 - 2i)$.

Step 1. Both factors are binomials, so FOIL, then convert the i^2 term with $i^2 = -1$ before combining.

Step 2. $3 - 6i + 4i - 8i^2 = 3 - 2i + 8$, so the product is **11 - 2i**.

Try it: Multiply $(2 + 5i)(4 - 3i)$.

check: $23 + 14i$

Skill 2 — Dividing with the conjugate

Rule: the conjugate of $c + di$ is $c - di$ — flip only the sign in front of i . Multiplying a conjugate pair gives $(c + di)(c - di) = c^2 + d^2$, a **sum** of squares (never $c^2 - d^2$), because $-d^2 i^2 = +d^2$. To divide, multiply the top *and* the bottom by the conjugate of the bottom.

Example: Simplify $\frac{7 + i}{1 - 2i}$.

Step 1. The bottom is not real, so multiply top and bottom by its conjugate $1 + 2i$ to clear the i downstairs.

Step 2. Top: $(7 + i)(1 + 2i) = 7 + 14i + i - 2 = 5 + 15i$. Bottom: $1^2 + 2^2 = 5$. Dividing each part by 5 gives **1 + 3i**.

Try it: Simplify $\frac{4 + 7i}{2 + i}$.

check: $3 + 2i$

Skill 3 — Non-real solutions come in pairs

Rule: if a quadratic (or any polynomial) with real coefficients has the non-real solution $a + bi$, then $a - bi$ is a solution too. A negative discriminant is the signal: $\sqrt{-k} = i\sqrt{k}$, and the $\pm$ in the quadratic formula produces the conjugate pair.

Example: Solve $x^2 - 4x + 13 = 0$.

Step 1. The left side has no real factor pair, so use the quadratic formula and watch the discriminant's sign.

Step 2. $16 - 52 = -36$, so $\sqrt{-36} = 6i$ and $x = \frac{4 \pm 6i}{2}$, giving $2 + 3i$ and $2 - 3i$.

Try it: Solve $x^2 + 2x + 10 = 0$.

check: $2 + 3i$ and $2 - 3i$

(!) Watch out: A conjugate changes ONLY the sign of the imaginary part: the conjugate of $7 - 3i$ is $7 + 3i$, not $-7 - 3i$ and not $7 - 3i$ again. And $(c + di)(c - di) = c^2 + d^2$ — if you ever write $c^2 - d^2$, you have treated i^2 as $+1$.